Effects extends Base

$$
\begin{array}{rcl}
\tau & ::= & \ldots \\
& | & \alpha^k \mid c^k \tau \ldots \mid \tau \to^\varepsilon \tau \mid \forall \alpha^k . \tau \\
& | & \texttt{blab} \\
\\
k & ::= & * \\
& | & k \to k \\
& | & \texttt{eff} \\
& | & \texttt{lab} \\
\\
m & ::= & d \\
& | & s \\
\\
e & ::= & \ldots \\
& | & \lambda^\varepsilon x : \tau . e \\
& | & e[\tau] \\
& | & \texttt{handle}_m(h, e) \\
& | & \texttt{error}_p \\
\\
h & ::= & \{\texttt{op} \to f, \ldots, \} \\
\\
f & ::= & \Lambda \alpha . \lambda^\varepsilon x : \tau_1 . \lambda^\varepsilon k : (\tau_2 \to^\varepsilon \tau) . e
\end{array}
\qquad
\begin{array}{rcl}
l & ::= & \text{Error Label} \\
\\
v & ::= & \ldots \\
& | & \lambda^\varepsilon x : \tau . e \\
& | & \Lambda \alpha^k . v \\
& | & \texttt{handler}^\varepsilon_m(h) \\
& | & \texttt{perform}^\varepsilon\ \texttt{op}\ \tau \\
& | & \texttt{error}_l \\
\\
E & ::= & \ldots \\
& | & E[\tau] \\
& | & \texttt{handle}^\varepsilon_m(h, E) \\
\\
F & ::= & \square \\
& | & F + e \mid v + F \mid F - e \mid v - F \\
& | & F \wedge e \mid v \wedge F \mid F \vee e \mid v \vee F \\
& | & F\ e \mid v\ F \\
& | & \text{if } F \text{ then } e \text{ else } e \\
& | & F[\tau]
\end{array}
$$

Listing 1: **Effects** language